

SAI VIKRANTH KANURU

kanuruvikranth@gmail.com | +91 9398596589 | Portfolio | GitHub | LinkedIn

Career Objective

B.Tech Electronics and Communication Engineering student building Generative AI and agentic systems, particularly Retrieval-Augmented Generation (RAG) applications. Passionate about solving problems end-to-end and developing production-grade AI solutions. Seeking an internship opportunity to contribute to real-world AI projects and collaborate with engineering teams.

Technical Skills

Programming: Python, C, SQL (MySQL, PostgreSQL), HTML, CSS

Frameworks & Libraries: LangChain, LangGraph, HuggingFace, FastAPI, Streamlit, Playwright, FastMCP

Databases & Retrieval: PostgreSQL, MySQL, Redis, Qdrant, FAISS, Chroma

AI/ML Concepts: LLMs, Embeddings, RAG, Prompt Engineering, Speech-to-Text (STT), Text-to-Speech (TTS), WebRTC

Tools: Docker, LangSmith, GitHub

Projects

Conversational Text2SQL Assistant & Obsidian Console

- Built a natural language SQL assistant with intelligent routing and self-healing query execution.
- Added SQL security layers to block destructive queries and mask sensitive data.
- Reduced query latency by 60% using Redis caching, connection pooling, and LangSmith tracing.
- Developed a console with live statistics, exports, and MySQL-to-PostgreSQL dialect conversion.

Multi-Source RAG AI Assistant

- Built a document assistant supporting PDFs, websites, text files, and scanned documents using Llama-Parse OCR.
- Improved retrieval accuracy by 25% through hybrid search, embeddings, and neural reranking.
- Reduced response time by 40% using LangGraph workflows with PostgreSQL memory.
- Built telemetry dashboards with retrieval metrics, audit trails, and LangSmith integration.

Multi-Threaded WebRTC Voice Agent

- Built a sub-100ms voice assistant using LiveKit WebRTC, Silero VAD, Deepgram STT, LLaMA 3.3, and Edge TTS.
- Added voice-controlled browser automation with LangGraph workflows for website interaction and app triggers.
- Achieved sub-2ms memory retrieval using PostgreSQL memory and Redis caching.
- Built a real-time 3D holographic UI with synchronized WebRTC data streams.

Education

Bachelor of Technology – Electronics and Communication Engineering

MVGR College of Engineering, Vizianagaram

Expected May 2028

CGPA: 7.06

Intermediate – Science (MPC)

Narayana Junior College, Vizianagaram

May 2024

CGPA: 8.72

Class 10 – SSC

Fort City School, Vizianagaram

March 2022

CGPA: 8.58

Certifications

- Python Essentials 1 & 2 – Cisco Networking Academy
- Quantum Foundations – Andhra Pradesh Government

Additional Information

Areas of Interest: Generative AI, Information Retrieval, Agentic Systems, System Design, DSA

Strengths: Problem-solving, analytical thinking, system-oriented approach, rapid learning, adaptability, and teamwork.